Joeri Bordes, PhD
Postdoc at University of Geneva, Switzerland

Neuroscientist using preclinical models to investigate stress-related disorders using machine learning for deep phenotyping of individual and social behaviors

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Publications: Google Scholar
Languages: Dutch (native), English, German

PROFESIONAL EXPERIENCE

Postdoctoral researcher, advised by Prof. Dr. Camilla Bellone — Since 01/2026
University of Geneva, Geneva, Switzerland

Visiting researcher with Prof. Dr. Marianne Müller — Since 02/2025
Leibniz Institute for Resilience Research, Mainz, Germany

Postdoctoral researcher, advised by Prof. Dr. Anna Beyeler — 10/2023–12/2025
Neurocentre Magendie U1215, INSERM, Bordeaux, France

PhD degree (Summa cum laude), advised by PD. Dr. Mathias Schmidt — 10/2018–11/2023
Max Planck Institute of Psychiatry, Munich, Germany

Research Intern, supervisor PD. Dr. Mathias Schmidt — 09/2017–05/2018
Max Planck Institute of Psychiatry, Munich, Germany

Research Intern, supervisor Prof. Dr. Ronald van Kesteren — 02/2017–08/2017
Free University, Center for Neurogenomics & Cognitive Research, Amsterdam, the Netherlands

Research Intern, supervisor Dr. Harm Krugers — 02/2016–07/2016
University of Amsterdam, Swammerdam Institute for Life Sciences, Amsterdam, the Netherlands

PRIZES AND AWARDS

- Junior Investigator Awardee for the Global Stress and Resilience Meeting, Lausanne, CH (link) 02/2027
- Travel awardee for the 35th IBNS Annual Meeting, Cape Town, SA (link) 02/2026
- Junior Investigator Awardee for the 7th Annual Swiss Stress Meeting, Bern, CH (link) 02/2025
- Best poster award for the 10th International Symposium on Resilience Research, Mainz, DE 09/2024
- Awarded highest grade (Summa cum laude) for PhD thesis and oral exam 11/2023
- Selected speaker for the ECNP Conference, Vienna, AT 10/2023
- Selected speaker at the 3rd Munich Winter Conference on Stress, Munich, DE (link) 03/2022
- Travel grant awardee for the EBBS conference, Lausanne, CH 09/2021
- Travel grant awardee for the FENS 2020 Forum, [online meeting] 07/2020
- Selected for the Neuropsychopharmacology ECNP workshop, Nice, France [cancelled] 03/2020
- Travel grant awardee for the EBBS conference, Praque, CZ 09/2019

European College of Neuropsychopharmacology (ECNP), European Brain and Behaviour Society (EBBS), Federation of European Neurosciences (FENS), International Behavioral Neuroscience Society (IBNS)

PEER-REVIEWED PUBLICATIONS: **21 ARTICLES**

* equally contributing author

17 Research Articles (3 as 1st author) 1 submitted, 1 in preparation

21. **Bordes**, Stont, Aman, Bittar, Soulat …, Beyeler (2026). Insular Cortex Corticotropin-Releasing Factor Signaling Mediates Social Alterations after Early Life Stress. **In preparation**. *[This research highlights that early life stress reshapes naturalistic social behavior via insular cortex dynamics, revealing sex-dependent effects and nominating the insula as a circuit target for normalizing social function]*

20. **Bordes**, Stont, Bajaj, Chang, Ebert, Miranda, Schlegel, Reinhardt, …, 9 authors …, Beyeler, Gassen, Schmidt (2026) Loss of noradrenergic Fkbp5 disrupts social behavior and norepinephrine dynamics in the basolateral amygdala. **Submitted**. (Preprint link) *[By linking Noradrenergic Fkbp5 regulation to phasic basolateral amygdala norepinephrine and mitochondrial/synaptic remodeling, this study highlights a circuit-specific route to normalize social salience without broadly suppressing noradrenergic function]*

19. Couderc, Dhani Reddy, Garg, Ricci, Vardiero, d'Almeida, Nicolas, **Bordes**, …, 6 authors …, Beyeler, (2026) Dopamine D1 receptor activation shapes anterior insula neural coding of anxiety. **Submitted**. (Preprint link) *[This study identifies a causal anterior insula D1 receptor pathway through which dopamine promotes anxiety and reshapes neural coding of exposed versus protected environments]*

18. Mitra, …, 10 authors …, **Bordes** …, 10 authors …, Schmidt, (2026) Deep Phenotyping with Global Brain Activity and Plasticity Mapping Identify the Dorsal Raphe-Basolateral Amygdala Circuit as a Mediator of Adaptive Stress Responses. **Submitted**. (Preprint link) *[This study combines social behavioral phenotyping with multimodal whole-brain imaging to identify system-wide network signatures of stress-coping trajectories and establish the dorsal raphe–basolateral amygdala circuit as a causal gatekeeper of adaptive stress responses]*

17. Bordes*, Ji*, Gasperoni, Sudre-Chinsky, Kalbasi, Harbich, Flachskamm, Fontanet, Narayan, Uhr, Namendorf, Bellone, Chen, Hausch, Lopez, Schmidt (2026). Pharmacological Inhibition of FKBP51 Mitigates Early Life

Adversity-Induced Social Deficits. Advanced Science (link) *[Early-life adversity disrupts social behavior and brain gene expression, which SAFit2 (an FKBP51 antagonist) can largely rescue]*

16. Narayan, Beer, Castoldi, Stark, Dal Bianco, Röh, Sauer, **Bordes**, …, 9 authors…, Binder, Schmidt (2026) Multilevel sex-influenced neurobiological signatures of early life adversity. PNAS (link) *[This study integrates whole-brain activity mapping, MRI, transcriptomics, and advanced behavioral phenotyping to reveal lasting, sex-specific effects of early life stress on adult brain function and behavior]*
15. De Donno*, Lopez*, Luecken*, Kos, Brivio, **Bordes**, Yang, Deussing, Schmidt, Theis, Chen (2026) Single-cell characterization of the adult male hippocampus suggests a prominent, and cell-type specific, role for Nrgn and Sgk1 in response to a social stressor. Molecular Psychiatry (link) *[Using large-scale single-cell RNA-seq of the adult male mouse posterior hippocampus under baseline vs. acute social defeat stress—plus GR/MR conditional knockouts in glutamatergic or GABAergic neurons—this study maps cell-type–specific stress transcriptional programs]*
14. Yang, Narayan, **Bordes**, …, 11 authors…, Lopez, Schmidt (2025) Mineralocorticoid receptor in glutamatergic neurons modulates anxiety exclusively in male mice via regulation of the actin bundling factor FAM107a. Biological Psychiatry Global Open Science (link) *[This work defines a cell-type–specific Mineralocorticoid Receptor pathway for stress-related behavior: male-selective baseline anxiety, associated hippocampal alterations, and a causal Fam107a mechanism capable of reversing the behavioral deficit]*
13. van Doeselaar, Abromeit, Stark, Menegaz, Mitra, Yang, Rehawi, Huettl, **Bordes**, …, 8 authors …, Schmidt (2025) FKBP51 in glutamatergic forebrain neurons promotes early life stress inoculation in female mice. Nature Communications (link) [*This research article highlights the importance of cell-type specific genetic stress risk markers on the early life stressed-induced behavioral alterations and brain genetic profile*]
12. **Bordes**, Bajaj, Miranda, …, 8 authors …, Gassen, Schmidt (2024) Sex-specific fear acquisition following early life stress is linked to amygdala and hippocampal purine and glutamate metabolism. Communications Biology (link) *[This research article shows that early life stress disrupts the HPA axis and machine-learning identified fear memory in a sex-specific manner, revealing differences in brain metabolism]*
11. Kovarova, **Bordes**, Mitra, Narayan, Springer, Brix, Deussing, Schmidt (2024) Deep phenotyping reveals CRH and FKBP51-dependent behavioral profiles following chronic social stress exposure in male mice. Neuropsychopharmacology (link) *[This research article highlights the importance of genetic stress risk markers on the social behavioral profile following chronic stress exposure]*
10. Miranda, **Bordes**, Pütz, Schmidt, Müller-Myhsok (2023) DeepOF: a Python package for supervised and unsupervised pattern recognition in mice motion tracking data. Journal of Open Source Software (link) *[This research article presents "DeepOF," an open-source Python package designed to analyze social behavior using AI and machine learning tools for both supervised and unsupervised classification]*
9. **Bordes***, Miranda*, Reinhardt, …, 11 authors …, Müller-Myhsok, Schmidt (2023) Automatically annotated motion tracking identifies a distinct social behavioral profile following chronic social defeat stress. Nature Communications (link) *[This research article utilizes the DeepOF open-source Python package, which employs machine learning to automate motion tracking and analyze stress-induced social behavior in freely interacting mice, enhancing precision and standardization in behavioral classification. The study demonstrates that DeepOF analysis reveals a more pronounced and robust stress-induced social behavioral phenotype]*
8. Kos, Lopez, **Bordes**, …, 13 authors …, Schmidt, Chen (2023) Early life adversity shapes social subordination and cell type–specific transcriptomic patterning in the ventral hippocampus. Science Advances (link) *[This research article employs advanced behavioral analysis tools and single-cell sequencing to investigate social behavior and hierarchy]*
7. van Doeselaar, Stark, Mitra, Yang, **Bordes**, …, 7 authors …, Lopez, Czisch, Schmidt (2023) Sex-specific and opposed effects of FKBP51 in glutamatergic and GABAergic neurons: Implications for stress susceptibility and resilience. PNAS (link) *[This research article emphasizes the impact of early life stress on cognitive function and brain structure volume, as assessed through mouse MRI]*
6. Brix, Monleon, Collado, Ederveen, Toksöz, **Bordes**, van Doeselaar, Engelhardt, Mitra, Narayan, Schmidt (2023) Metabolic effects of early life stress and pre-pregnancy obesity are longlasting and sex-specific in mice. European Journal of Neuroscience (link) *[This research article finds that early life stress affects male body weight long-term, while females adapt, possibly by stabilizing their microbiota]*
5. Brix, Toksöz, Aman, Kovarova, Springer, **Bordes**, …, 6 authors …, Deussing, Schmidt (2022) Contribution of the co-chaperone FKBP51 in the ventromedial hypothalamus to metabolic homeostasis in male and female mice. Molecular Metabolism (link) *[This research article shows that FKBP51 manipulation in SF1-expressing VMH cells has a mild impact on metabolism, indicating its subsidiary role compared to broader MBH-wide changes]*
4. Brix, Häusl, Toksöz, **Bordes**, …, 6 authors …, Chen, Schmidt (2022) The co-chaperone FKBP51 modulates HPA axis activity and age-related maladaptation of the stress system in pituitary proopiomelanocortin cells. Psychoneuroendocrinology (link) *[This research article shows that FKBP51 deficiency in POMC-expressing pituitary cells enhances HPA axis negative feedback and protects against age-related disruptions in corticosterone rhythms]*
3. Engelhardt, Tang, Elkhateib, **Bordes**, …, 7 authors …, Deussing, Schmidt (2021) FKBP51 in the Oval Bed Nucleus of the Stria Terminalis Regulates Anxiety-Like Behavior. eNeuro (link) *[This research article underscores the impact of genetic risk factors on the development of stress-induced anxiety-like behavioral symptoms]*

2. Bonapersona, Hoijtink, RELACS Consortium: (Abbinck, Baram, Bolton, **Bordes**, ..., 12 authors), Sarabdjitsingh, Joëls (2021) Increasing the statistical power of animal experiments with historical control data. Nature Neuroscience (link) *[This research article illustrates the benefits of utilizing historical control data to minimize the number of animals used and improve statistical power in studies on early life stress-induced behavioral symptoms]*

1. van Doeselaar, Yang, **Bordes**, Brix, Engelhardt, Tang & Schmidt (2020) Chronic social defeat stress in female mice leads to sex-specific behavioral and neuroendocrine effects. Stress (link) *[This research article showed the importance of sex in studying the effects of chronic stress exposure]*

7 Review, commentary & book chapters (3 as 1st author)

7. Bordes, Ricci, Stont, Bittar, Beyeler (2027) Chapter 17. Decoding Anxiety Circuits by Integrating Behavioral Assays with Neural Recordings. Psychiatric Vulnerability, and Mood and Anxiety Disorders, (link) *[This book chapter describes how behavioral assays and neural recording techniques can be integrated to decode the neural circuits underlying anxiety, highlighting their potential to improve mechanistic understanding and translational relevance]*

6. Bittar, **Bordes**, Nicolas, Calhoon, Beyeler (2025) Chapter 30. Pre-Clinical Models of Emotional Dysregulations. Handbook of Human Affective Neuroscience, 2nd edition (link) *[This book chapter describes how animal models help elucidate emotional processing and psychiatric disorders, stressing the need for advanced computational tools to improve insights and treatment development]*

5. Albayrak, de Fátima da Silva Vaz, **Bordes**, Ünlü, Sep, Vinkers, Pinto, Yapıcı Eser (2024) Translational models of stress and resilience: An applied neuroscience methodology review Neuroscience Applied (link) *[This review describes the complexity of resilience to stress and explains how translational models, including cell cultures and rodents, are used to study stress and resilience mechanisms, highlighting the need for improved models]*

4. Voulgaropoulou, Bastiaanssen, Alves, Viglione, **Bordes**, Jurek, Paribello, Sep (2024) Editorial: An interdisciplinary perspective on resilience - A special section in Neuroscience Neuroscience Applied (link) *[This editorial outlines the mission of the Resilience Network from the European College of Neuropsychopharmacology, focusing on its interdisciplinary approach to resilience research through various key pillars]*

3. Bordes, Miranda, Müller-Myhsok, Schmidt (2023) Advancing social behavioral neuroscience by integrating ethology and comparative psychology methods through machine learning. Neuroscience & Biobehavioral Reviews (link) *[This review highlights the historical impact of behavioral neuroscience and emphasizes the crucial role of cutting-edge machine learning tools in computational neuroscience for advancing the analysis of social behavior in the context of stress exposure]*

2. Bordes*, Miranda*, Gasperoni, Lopez (2023) Increasing resolution in stress neurobiology: from single cells to complex group behaviors. Stress (link) *[This commentary highlights the latest molecular tools in stress neuroscience and underscores the importance of computational neuroscience in advancing this field]*

1. von Mücke-Heim, Urbina-Treviño, **Bordes**, Ries, Schmidt, Deussing (2023) Introducing a depression-like syndrome for translational neuropsychiatry: a plea for taxonomical validity and improved comparability between humans and mice. Molecular Psychiatry (link) *[This review highlights the difficulties in translating preclinical stress models to clinical settings, emphasizing the need for advanced computational tools and better alignment with clinical symptoms. This is relevant to the current research proposal, as similar issues affect preclinical models of anxiety disorders]*

COMMUNITY ACTIVITIES

• Vice-president of the Synapsy Academy at the University of Geneva, CH (link)	Since 2025
• Selected member of the ECNP Early Career Academy, founder of the Open Science ECNP group (link)	2025-2030
• Member of Diversity, Equity & Inclusion (DEI) Committee, Neurocentre Magendie, Bordeaux, FR	2024-2025
• Member of ALBA Network for Diversity in Neurosciences	Since 2024
• Active member of the European College of Neuropsychopharmacology Resilience Network (link)	Since 2023
• Member of the European Brain and Behaviour Society (EBBS)	Since 2018
• Member of the Federation of European Neurosciences (FENS)	Since 2018

ORGANISATION OF INTERNATIONAL CONFERENCES

2026-2028: Member of the scientific programme committee for the ECNP Conference 2028
2025-2026: Member of the 2026 Brain Conference, "BrainStress", Bordeaux, France (link)
2024-present: Member of the organization committee for the Resilience Symposium, Mainz, Germany (link)
2019-2022: Member of the local organizing committee for the 2nd and 3rd Munich Winter Conference on Stress (link)

SUPERVISING AND MENTORING ACTIVITIES

• Supervision and mentoring of 1 ongoing PhD thesis in Geneva, 2 MSc and 1 BSc theses in Bordeaux, and 3 MSc theses in Munich, all students have been included as co-authors on publications or manuscripts.

EDUCATION

PhD degree in Neuroscience (Summa cum laude, highest distinction)	10/2018–11/2023
Max Planck Institute of Psychiatry & Ludwig-Maximilians-University, Munich, DE	
Master of Science in Neuroscience (GPA: 8/10), University of Amsterdam, NL	09/2016–09/2018
Bachelor of Science in Psychobiology (GPA: 7/10), University of Amsterdam, NL	09/2012–09/2016